## Module 4: Core organic chemistry

This module introduces organic chemistry and its important applications to everyday life, including current environmental concerns associated with sustainability.

The module assumes knowledge and understanding of the chemical concepts developed in Module 2: Foundations in chemistry.

The module provides learners with a knowledge and understanding of the important chemical ideas that underpin the study of organic chemistry:

- nomenclature and formula representation, functional groups, organic reactions and isomerism
- aliphatic hydrocarbons
- alcohols and haloalkanes
- organic practical skills and organic synthesis
- instrumental analytical techniques to provide evidence of structural features in molecules.

This module also provides learners with an opportunity to develop important organic practical skills, including use of Quickfit apparatus for distillation, heating under reflux and purification of organic liquids.

In the context of this module, it is important that learners should appreciate the need to consider responsible use of organic chemicals in the environment. Current trends in this context include reducing demand for hydrocarbon fuels, processing plastic waste productively, and preventing use of ozone-depleting chemicals.

**Synoptic assessment**

This module provides a context for synoptic assessment and the subject content links strongly with the content encountered in Module 2: Foundations in chemistry.

- Atoms, moles and stoichiometry
- Acid and redox reactions
- Bonding and structure

Knowledge and understanding of Module 2 will be assumed and examination questions will be set that link its content with this module and other areas of chemistry.

## 4.1 Basic concepts and hydrocarbons

This section is fundamental to the study of organic chemistry.

This section introduces the various types of structures used routinely in organic chemistry, nomenclature, and the important concepts of homologous series, functional groups, isomerism and reaction mechanisms using curly arrows.

The initial ideas are then developed within the context of the hydrocarbons: alkanes and alkenes.

## 4.1.1 Basic concepts of organic chemistry

| Learning outcomes | Additional guidance |
|---|---|
| *Learners should be able to demonstrate and apply their knowledge and understanding of:* | |

### Naming and representing the formulae of organic compounds

**(a)** application of IUPAC rules of nomenclature for systematically naming organic compounds

Nomenclature will be limited to the functional groups within this specification.
E.g. $CH_3CH_2CH(CH_3)CH_2OH$ has the systematic name: 2-methylbutan-1-ol.
Learners will be expected to know the names of the first ten members of the alkanes homologous series and their corresponding alkyl groups.

HSW8 Use of systematic nomenclature to avoid ambiguity.

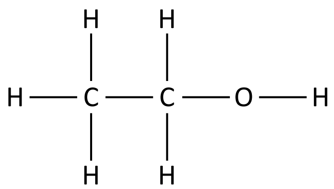
HSW11 The role of IUPAC in developing a systematic framework for chemical nomenclature.

**(b)** interpretation and use of the terms:

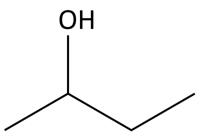
**(i)** *general formula* (the simplest algebraic formula of a member of a homologous series) e.g. for an alkane: $C_nH_{2n+2}$

**(ii)** *structural formula* (the minimal detail that shows the arrangement of atoms in a molecule) e.g. for butane: $CH_3CH_2CH_2CH_3$ or $CH_3(CH_2)_2CH_3$

**(iii)** *displayed formula* (the relative positioning of atoms and the bonds between them) e.g. for ethanol:



**(iv)** *skeletal formula* (the simplified organic formula, shown by removing hydrogen atoms from alkyl chains, leaving just a carbon skeleton and associated functional groups) e.g. for butan-2-ol:

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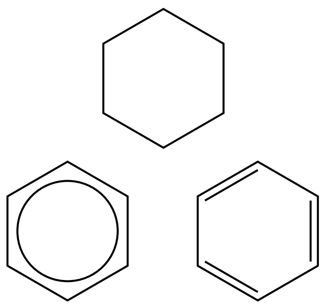
**See also 2.1.3 b** for empirical formula and molecular formula.

Definitions **not** required.

In structural formulae, the carboxyl group will be represented as COOH and the ester group as COO.

The symbols below will be used for cyclohexane and benzene:



HSW8 Communication using organic chemical structures; selecting the appropriate type of formula for the context.

**Functional groups**

**(c)** interpretation and use of the terms:

- **(i)** *homologous series* (a series of organic compounds having the same functional group but with each successive member differing by $CH_2$)
- **(ii)** *functional group* (a group of atoms responsible for the characteristic reactions of a compound)
- **(iii)** *alkyl group* (of formula $C_nH_{2n+1}$)
- **(iv)** *aliphatic* (a compound containing carbon and hydrogen joined together in straight chains, branched chains or non-aromatic rings)
- **(v)** *alicyclic* (an aliphatic compound arranged in non-aromatic rings with or without side chains)
- **(vi)** *aromatic* (a compound containing a benzene ring)
- **(vii)** *saturated* (single carbon–carbon bonds only) and *unsaturated* (the presence of multiple carbon–carbon bonds, including C=C, C≡C and aromatic rings)

Definition required for homologous series only.

R may be used to represent alkyl groups, but also other fragments of organic compounds not involved in reactions.

The terms saturated and unsaturated will be used to indicate the presence of multiple carbon–carbon bonds as distinct from the wider term 'degree of saturation' used also for any multiple bonds and cyclic compounds.

**(d)** use of the general formula of a homologous series to predict the formula of any member of the series

**Isomerism**

**(e)** explanation of the term *structural isomers* (compounds with the same molecular formula but different structural formulae) and determination of possible structural formulae of an organic molecule, given its molecular formula

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**Reaction mechanisms**

**(f)** the different types of covalent bond fission:

- **(i)** homolytic fission (in terms of each bonding atom receiving one electron from the bonded pair, forming two radicals)
- (ii) heterolytic fission (in terms of one bonding atom receiving both electrons from the bonded pair)

| | | |
|---|---|---|
| **(g)** | the term *radical* (a species with an unpaired electron) and use of 'dots' to represent species that are radicals in mechanisms | Radical mechanisms will be represented by a sequence of equations.<br>Dots, •, are required in all instances where there is a single unpaired electron (e.g. $Cl$• and $CH_3$•). Dots are **not** required for species that are diradicals (e.g. O). |
| **(h)** | a 'curly arrow' described as the movement of an electron pair, showing either heterolytic fission or formation of a covalent bond | 'Half curly arrows' are **not** required, **see 4.1.2 f**.<br><br>HSW1,8 Use of the 'curly arrow' model to demonstrate electron flow in organic reactions. |
| **(i)** | reaction mechanisms, using diagrams, to show clearly the movement of an electron pair with 'curly arrows' and relevant dipoles. | Any relevant dipoles should be included.<br>Curly arrows should start from a bond, a lone pair of electrons or a negative charge.<br><br>HSW1,2,8 Use of reaction mechanisms to explain organic reactions. |